

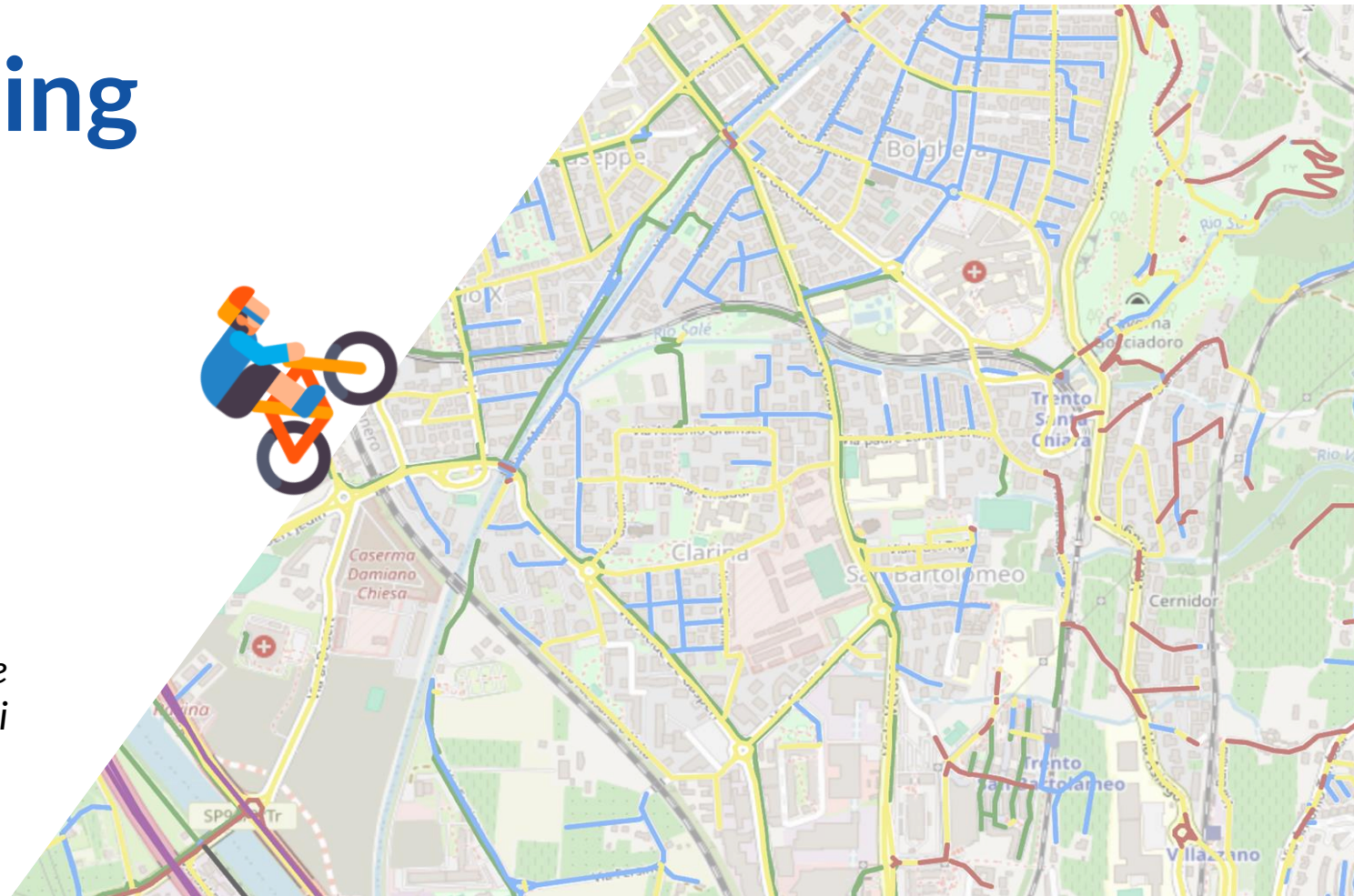
Level of Traffic Stress and Infrastructure Network Analysis for Safe and Accessible Cycling

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Convention Conference, Brdo pri
Kranju (SI), January 21st 2025



Background

We need to talk about
sustainability and bikes in
Alpine cities.



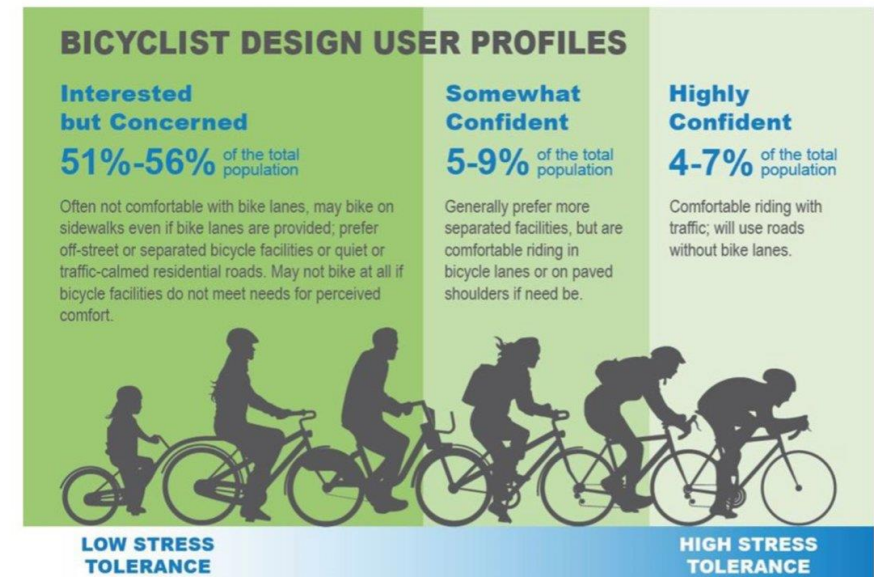
Background

How well the **bike network** in a community connects people with the places they want to go?



Background

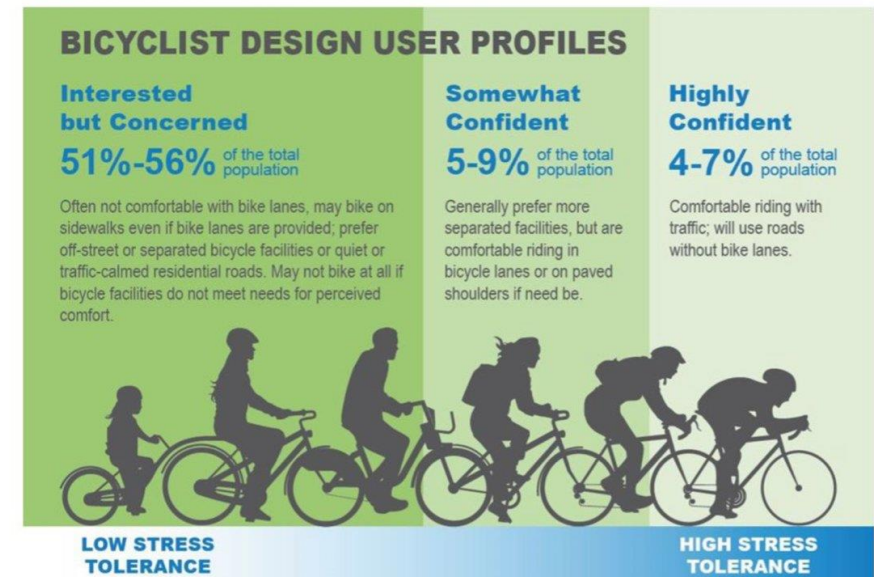
The **Traffic Stress** concept is a detailed metric used to evaluate *the range of difficulties and discomforts encountered by cyclists on specific road segments. It encompasses both the immediate feelings of discomfort, anxiety, or danger experienced by cyclists, as well as the wider effects of factors such as infrastructure design, vehicle behavior, environmental conditions, and topography**.



* Winters, M., Davidson, G., Kao, D., & Teschke, K. (2011). Motivators and deterrents of bicycling: comparing influences on decisions to ride. *Transportation*, 38, 153-168.

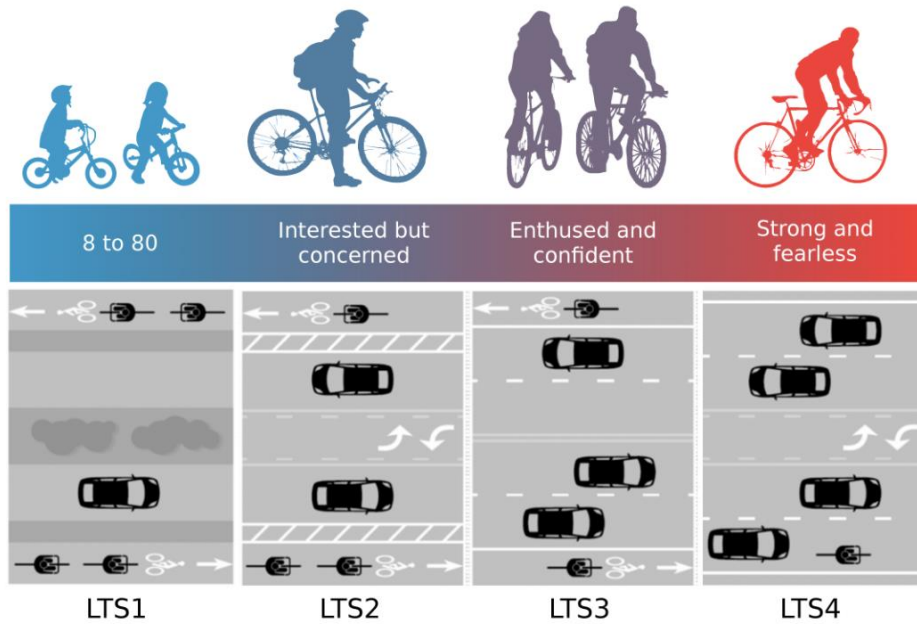
Background

The **Traffic Stress** concept is a detailed metric used to evaluate *the range of difficulties and discomforts encountered by cyclists on specific road segments*. It **encompasses** both the *immediate feelings of discomfort, anxiety, or danger experienced by cyclists*, as well as the *wider effects of factors such as infrastructure design, vehicle behavior, environmental conditions, and topography* *.



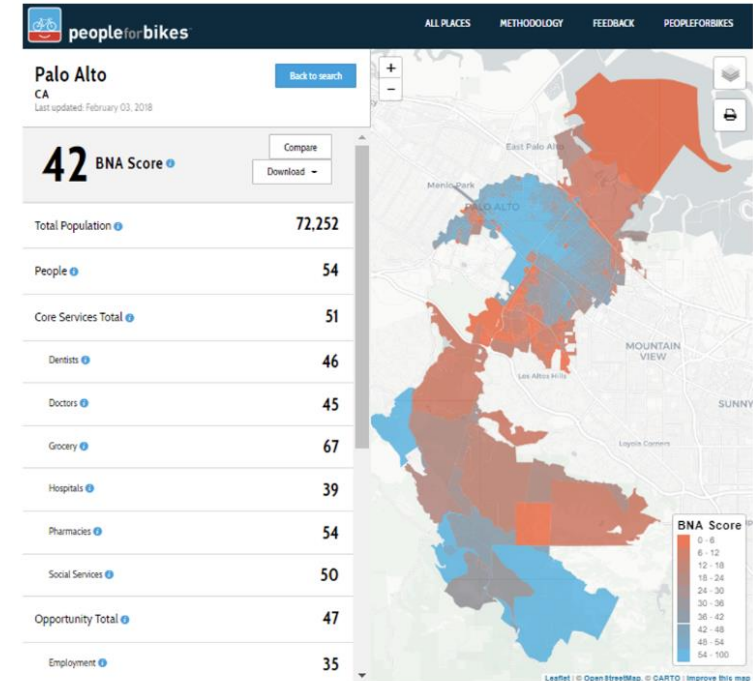
* Winters, M., Davidson, G., Kao, D., & Teschke, K. (2011). Motivators and deterrents of bicycling: comparing influences on decisions to ride. *Transportation*, 38, 153-168.

Background



Levels of Traffic Stress

Mekuria, M. C., Furth, P. G., & Nixon, H. (2012). Low-stress bicycling and network connectivity.



PeopleForBikes BNA score

PeopleForBikes. (2017). Methodology BNA. Retrieved from PeopleForBikes website: <https://bna.peopleforbikes.org/placesforbikes/methodology/>

Research Goals

1. Investigate the **factors influencing the choice between bicycles and cars**, integrating aspects such as logistical and environmental conditions.
2. Focus on **quantifying the relationship between traffic stress levels and risk perception**, considering infrastructural and environmental conditions (like centrality measures and network gaps) and qualitative factors (like accessibility and accident presence).
3. Develop a **Practical Open Digital Tool (LTS-Bike Plan)** to aid public administrations by providing insights for potential interventions to increase bicycle usage.
4. Utilize **Trento and Bolzano as case studies** due to familiarity with these areas for effective research validation and their **geomorphological variability (flat vs challenging terrains)**.

Research Questions

- 1) In urban landscapes with diverse geomorphological characteristics, **which streets or areas exhibit high LTS**, and which ones are characterized by low stress?
- 2) **Which streets and areas in the urban landscape exhibit higher BNA scores** and are near common destinations, including services and amenities?
- 3) Are the networks composed of subgraphs distinguished by different levels of traffic stress? If this is the case, **how many high-stress gaps occur within different areas of low stress?** Moreover, which of these high-stress gaps are most **crucial**?

One of the **future goals** of the work of Vybournova, A., Cunha, T., Gühnemann, A., & Szell, M. (2023). *Automated detection of missing links in bicycle networks*. *Geographical Analysis*, 55(2), 239-267.



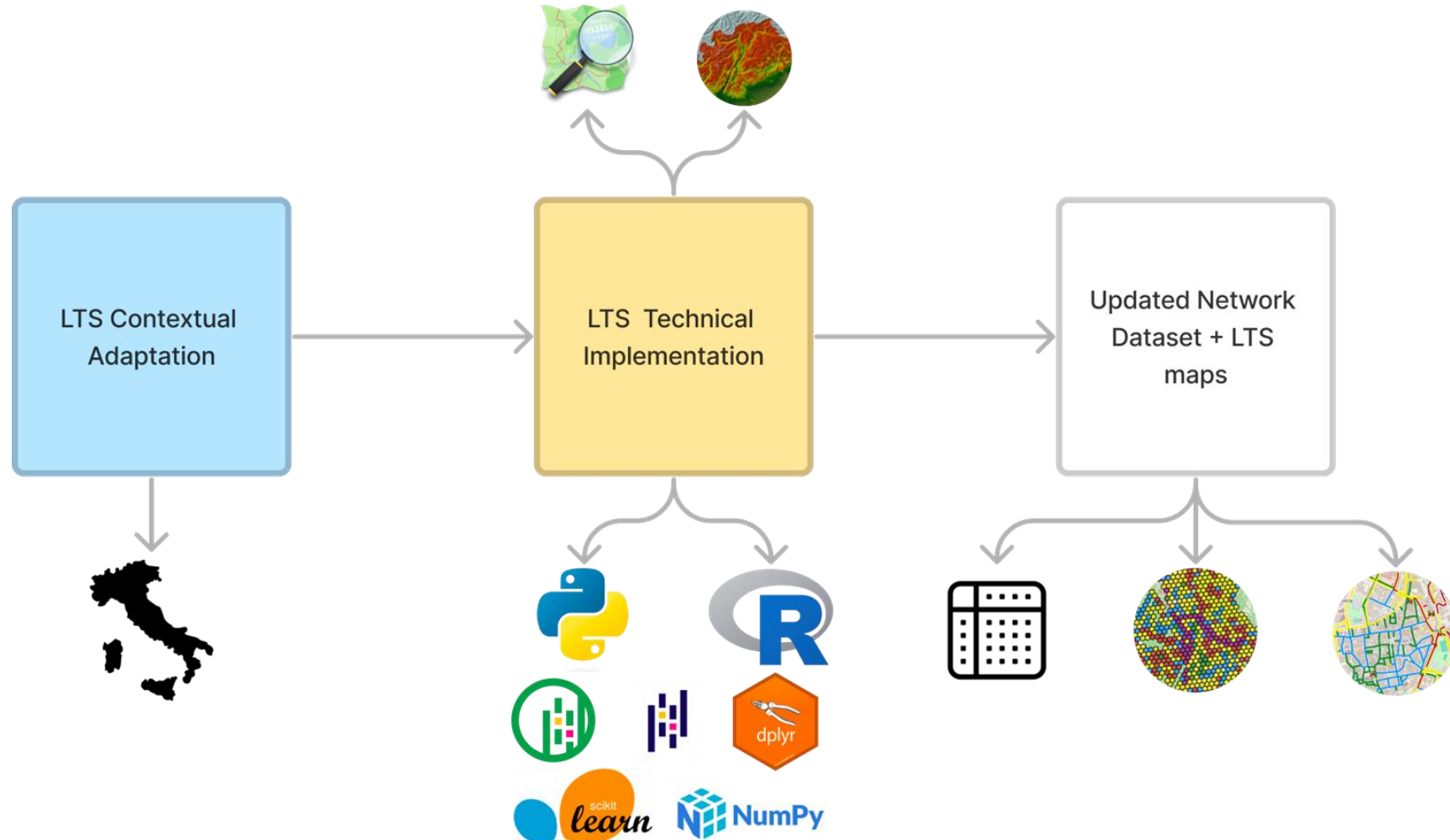
Research Questions

- 4) Does a **spatial relationship** exist between the **locations of accidents** and the **high traffic stress gaps** within the urban network?
- 5) Is it possible to **predict areas at high risk for accidents** by combining quantitative metrics, such as centrality measures, with qualitative assessments like accessibility scores and levels of traffic stress?



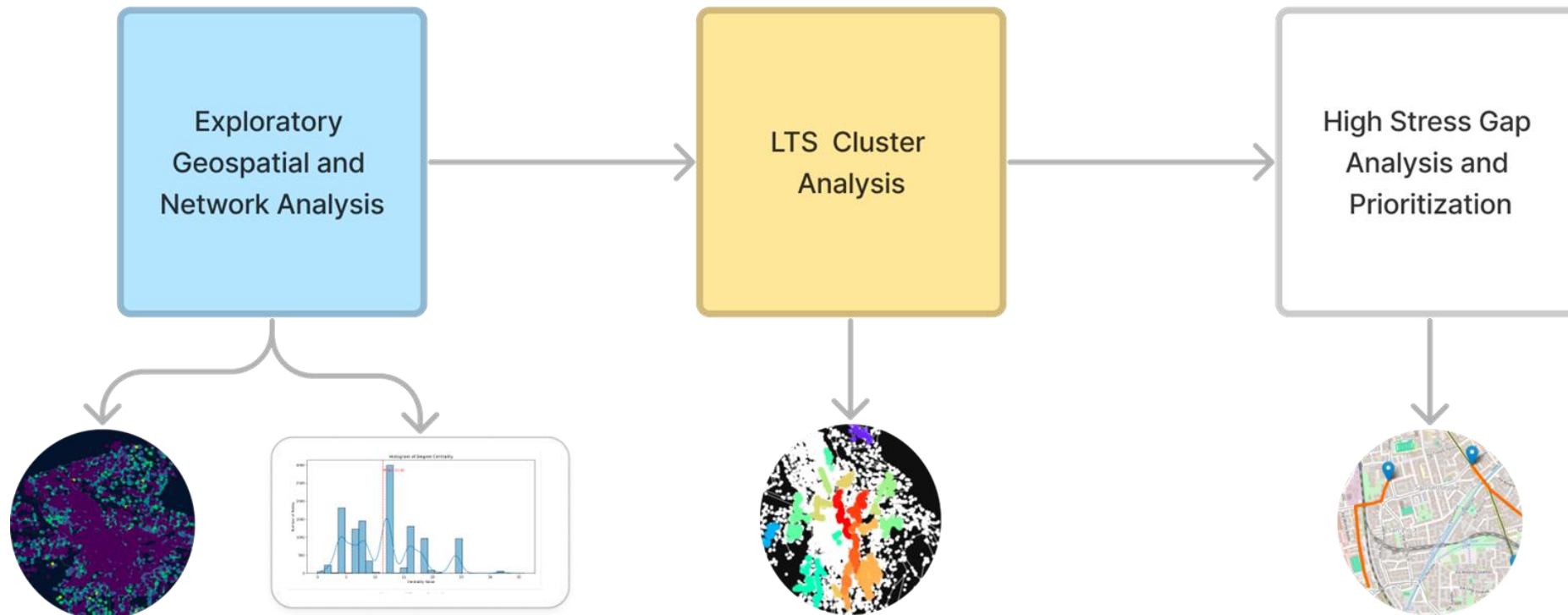
Methodology

Stage One: Adaptation of the LTS Scoring Process



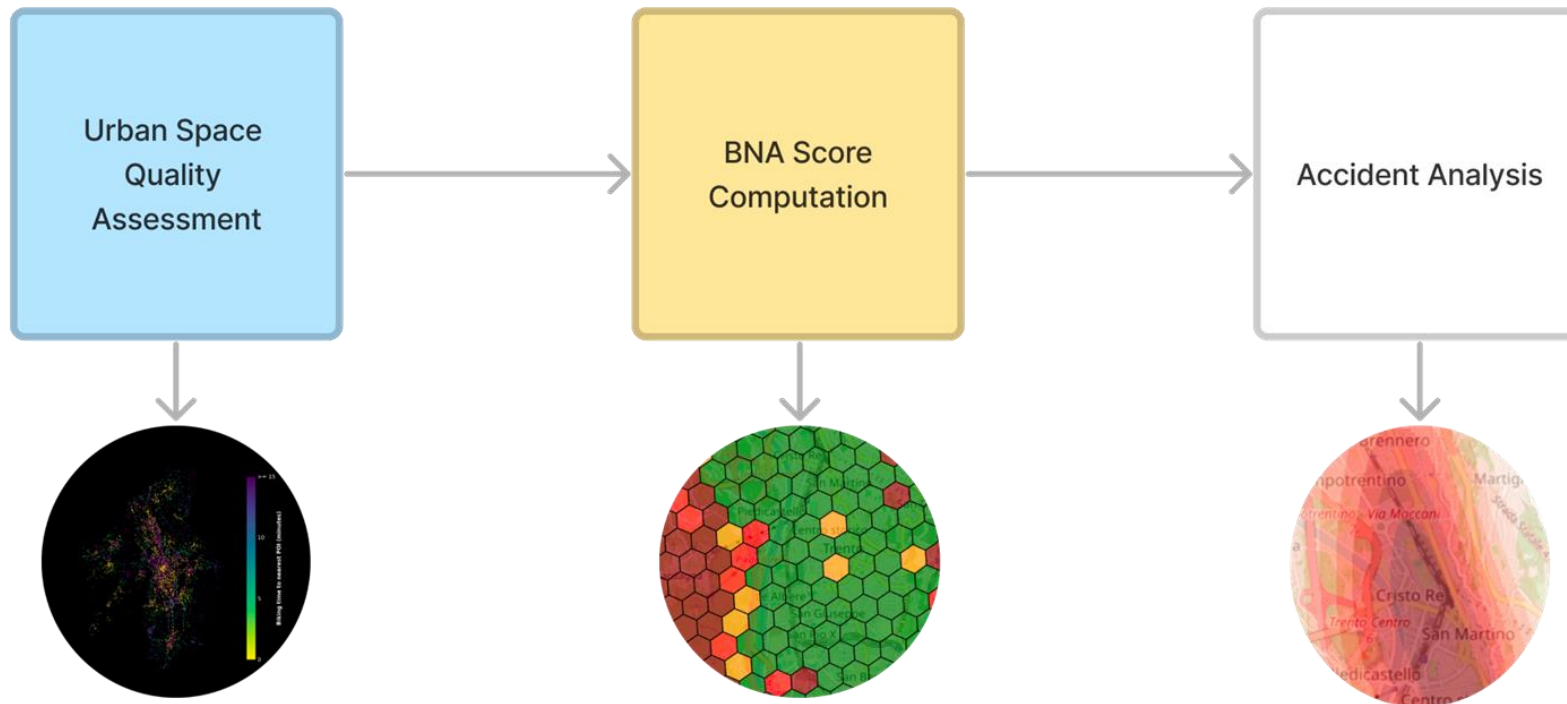
Methodology

Stage Two: Infrastructure and Network Analysis



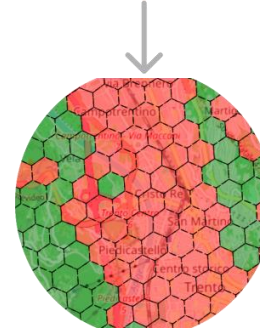
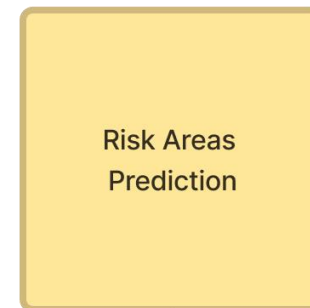
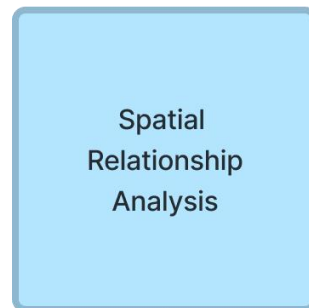
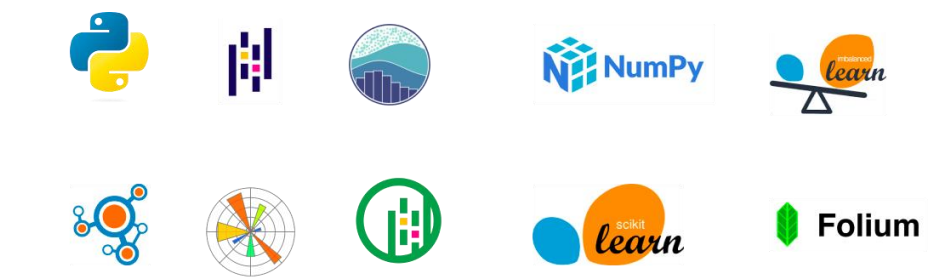
Methodology

Stage Three: Qualitative Analysis



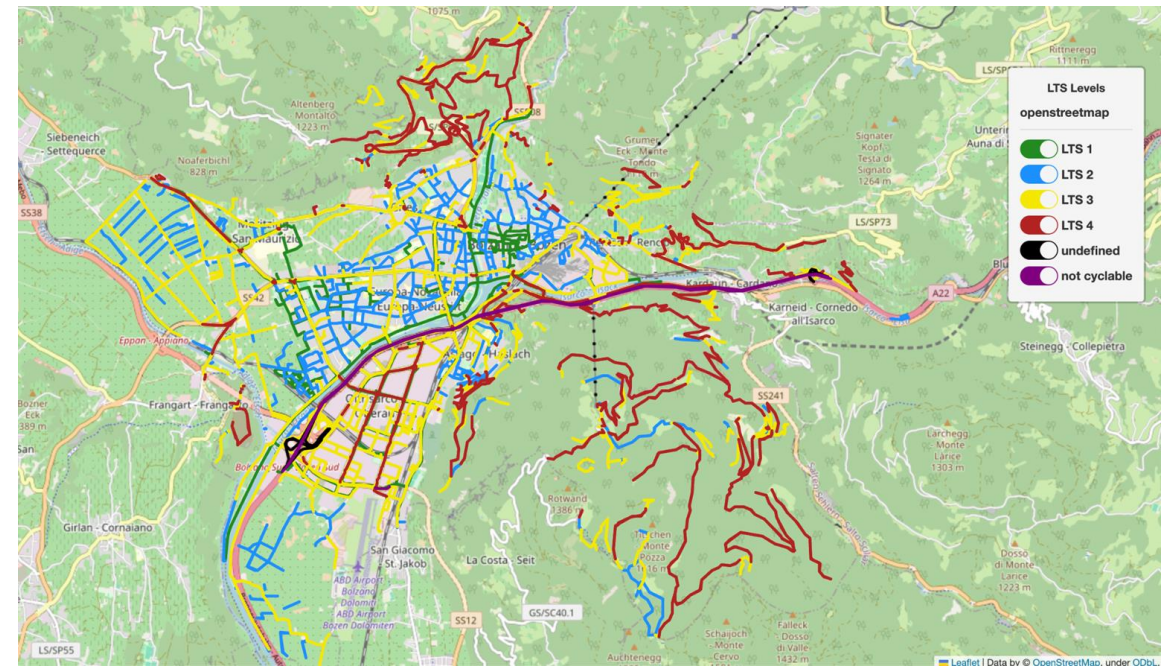
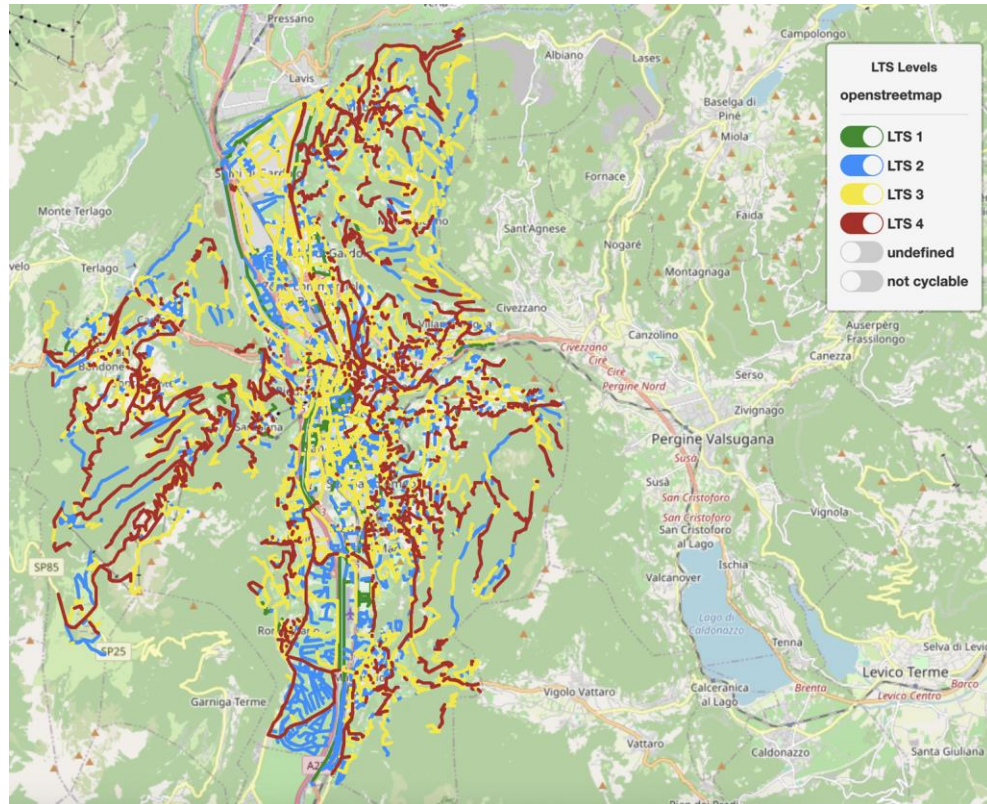
Methodology

Final Steps Analysis



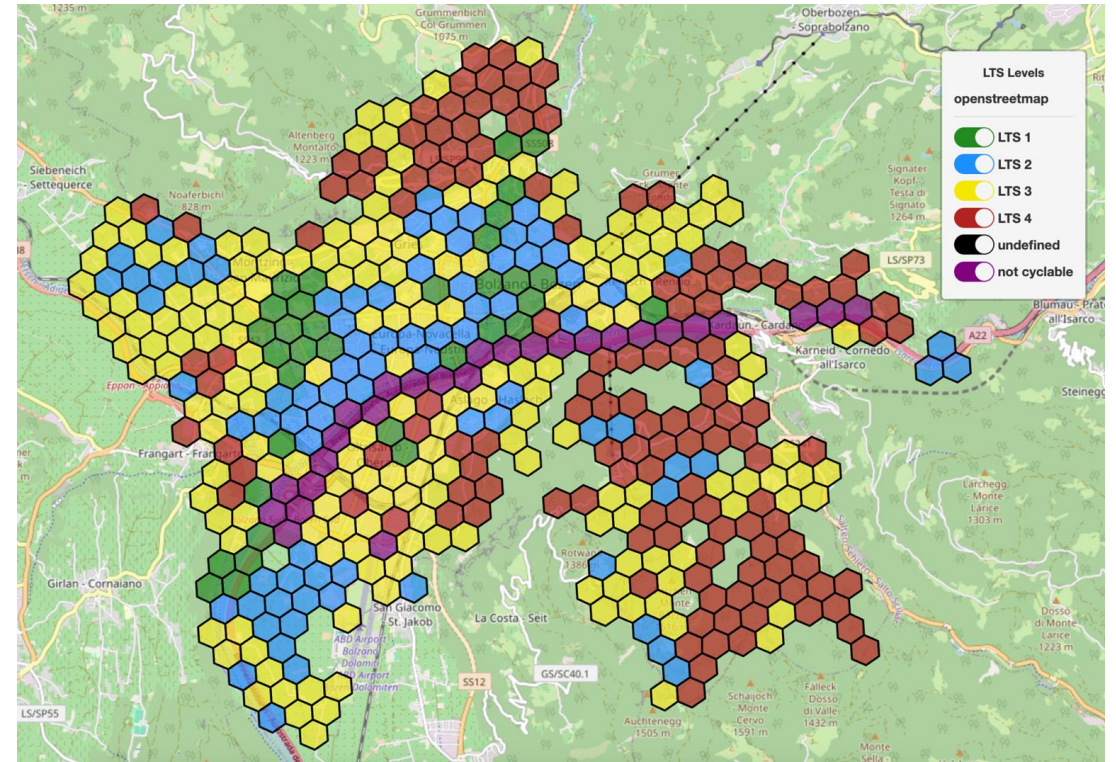
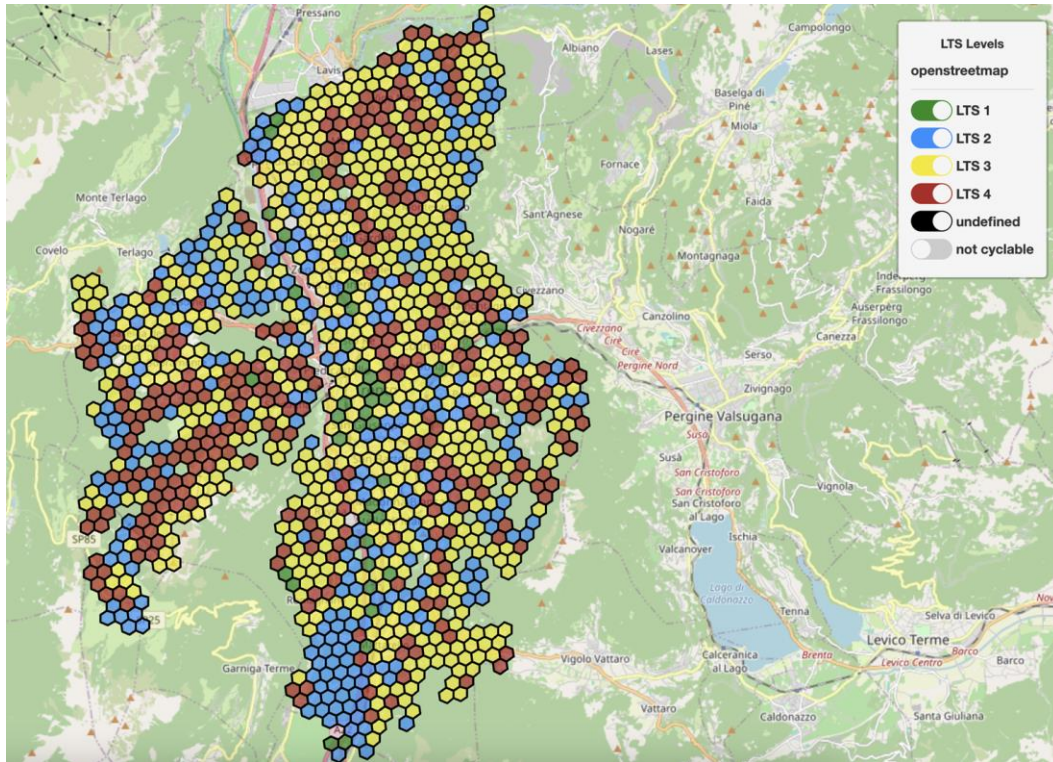
Results

Adaptation of the LTS Scoring Process (Trento and Bolzano) - LTS map



Results

Adaptation of the LTS Scoring Process (Trento and Bolzano) - Choropleth Map

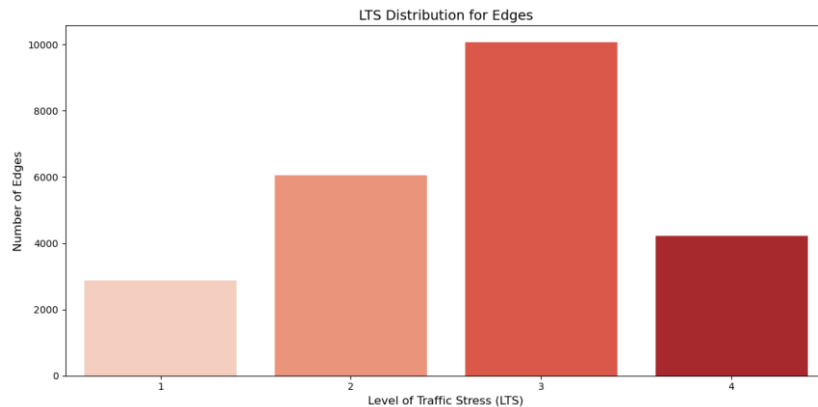


Grid of 400 m hexagonal areas called H3 (Uber Engineering. H3: Uber's Hexagonal Hierarchical Spatial Index. Accessed August 2023, from <https://eng.uber.com/h3/>) - each area is colored depending on the most prevalent LTS

Results

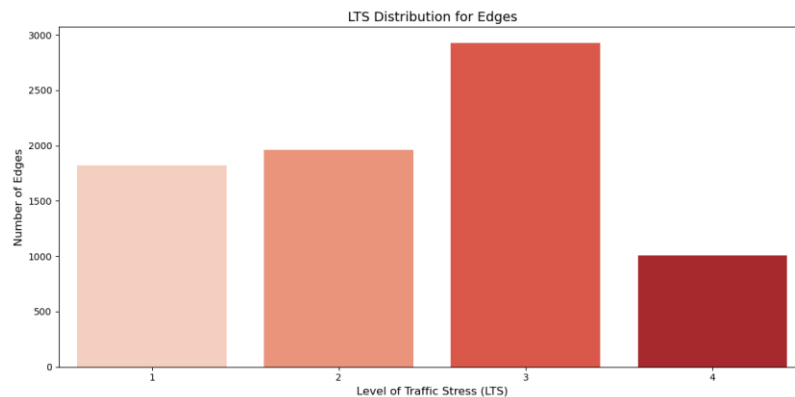
Infrastructure and Network Analysis - LTS Distribution

Trento



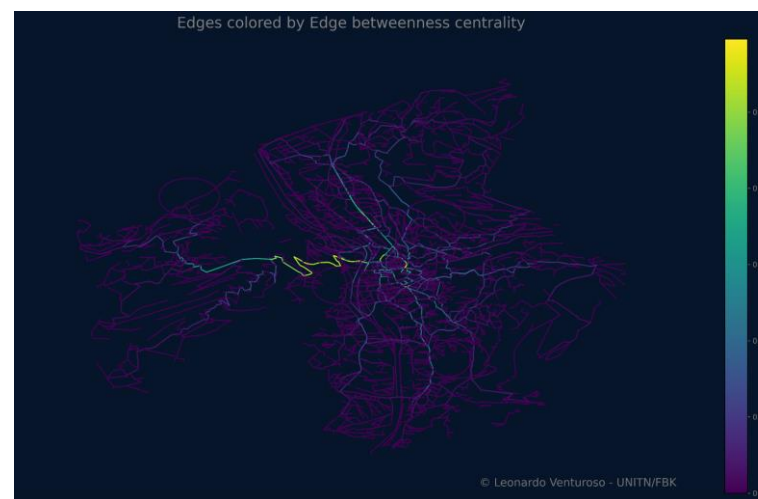
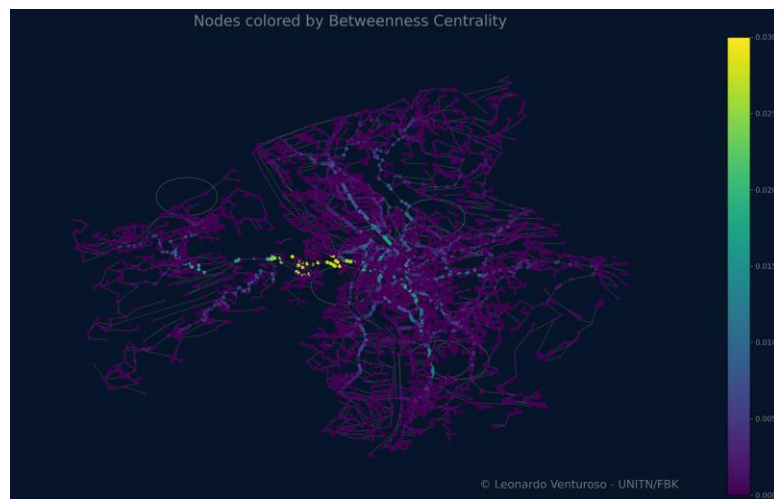
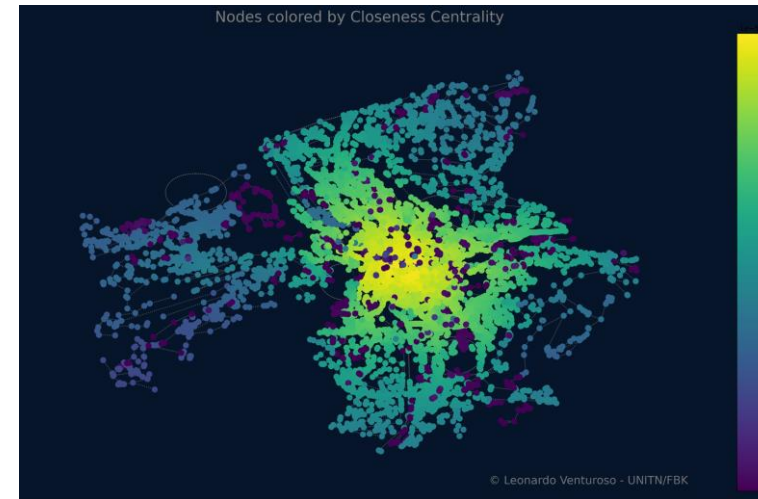
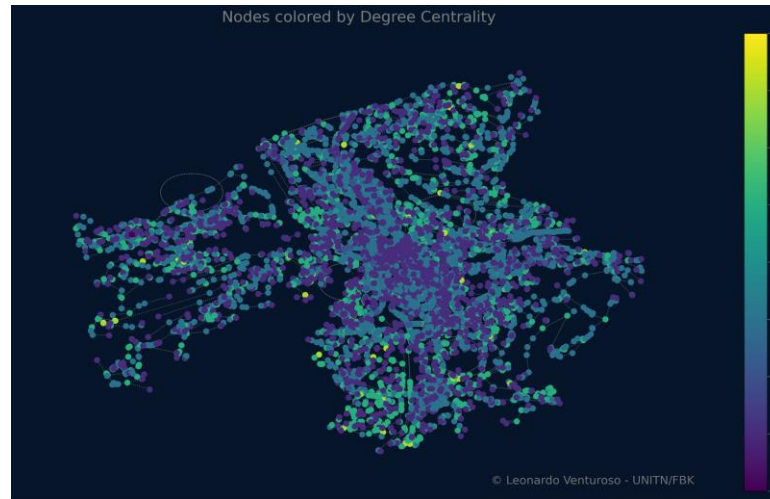
The examination of LTS level distributions showed comparable patterns across both case studies, with **LTS 3** emerging as the **predominant level**. Specifically in the case of Bolzano, it was observed that LTS 1 and 2 exhibit a notably similar presence.

Bolzano



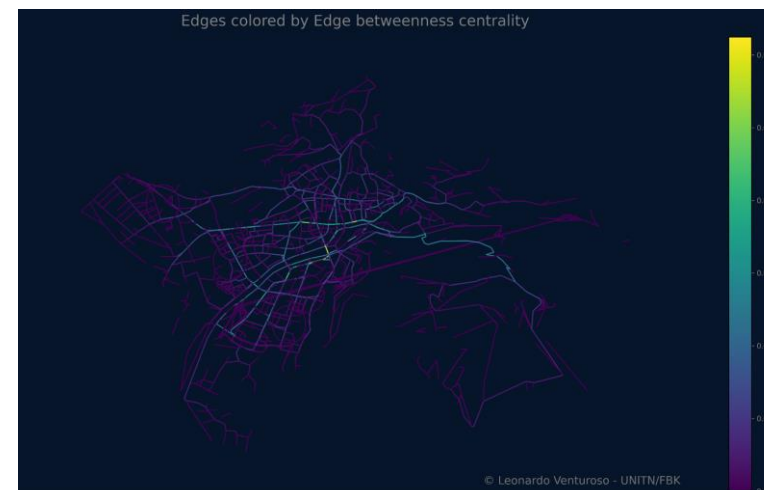
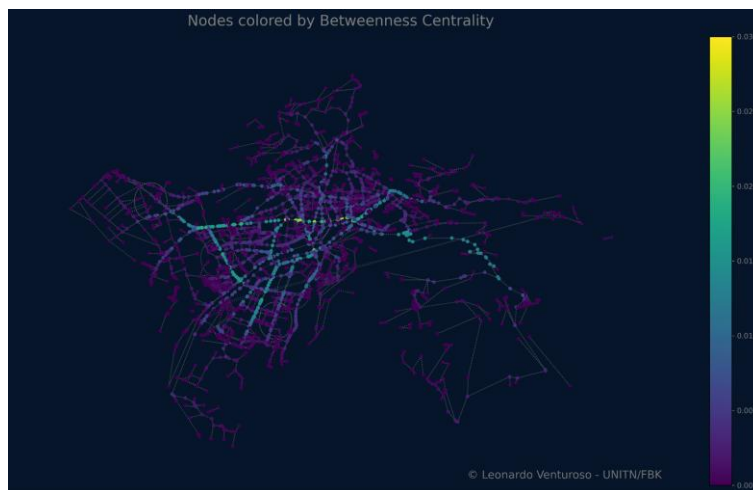
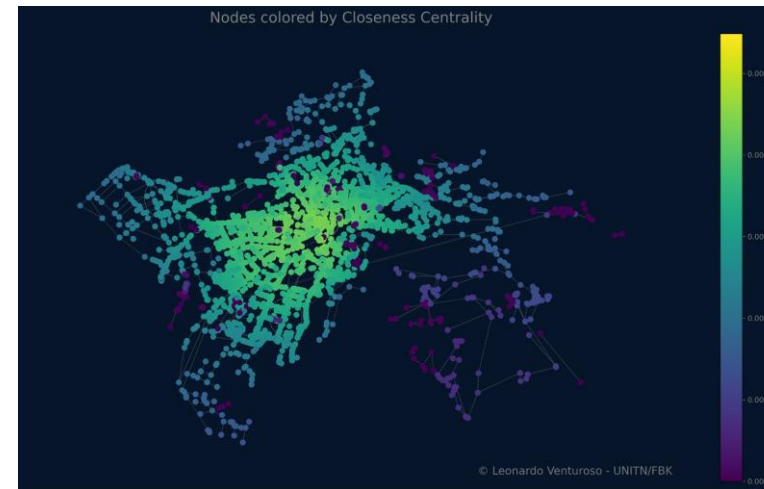
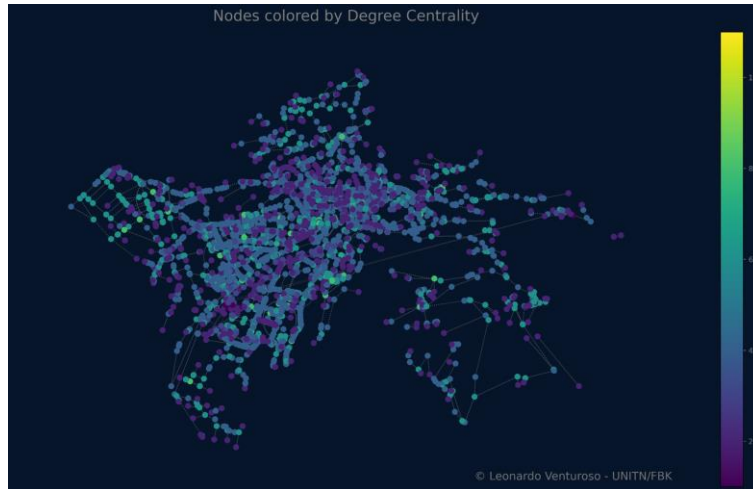
Results

Infrastructure and Network Analysis (Trento) - Centrality Measures



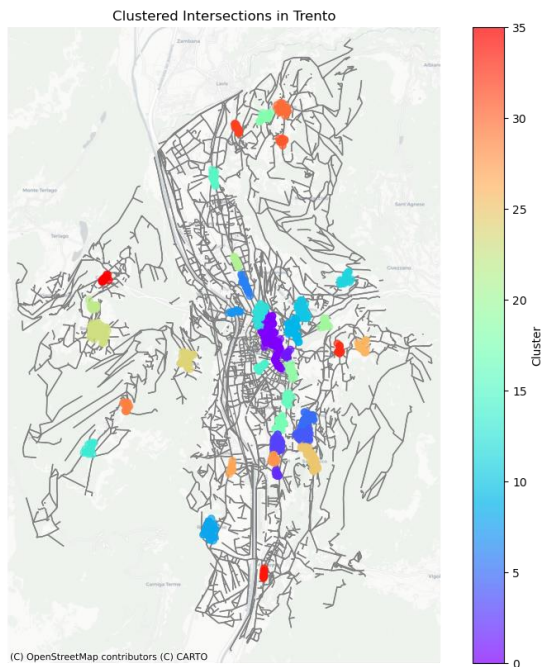
Results

Infrastructure and Network Analysis (Bolzano) - Centrality Measures

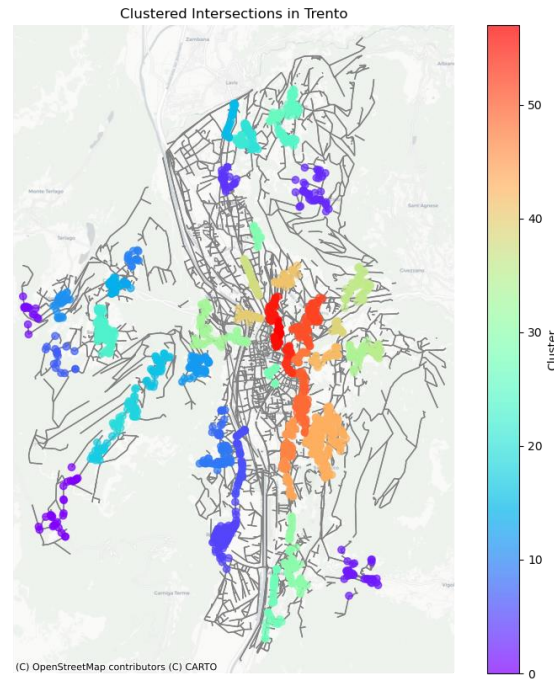


Results

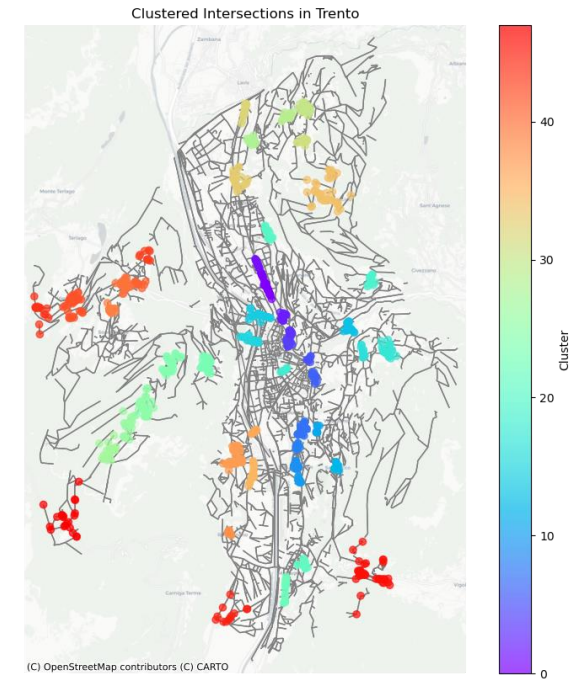
Infrastructure and Network Analysis (Trento) - Clustering High Stress (DBSCAN, HDBSCAN, OPTICS)



Silhouette Score: 0.62026
Davies-Bouldin Score: 0.37756
Calinski-Harabasz Score: 8799.21625



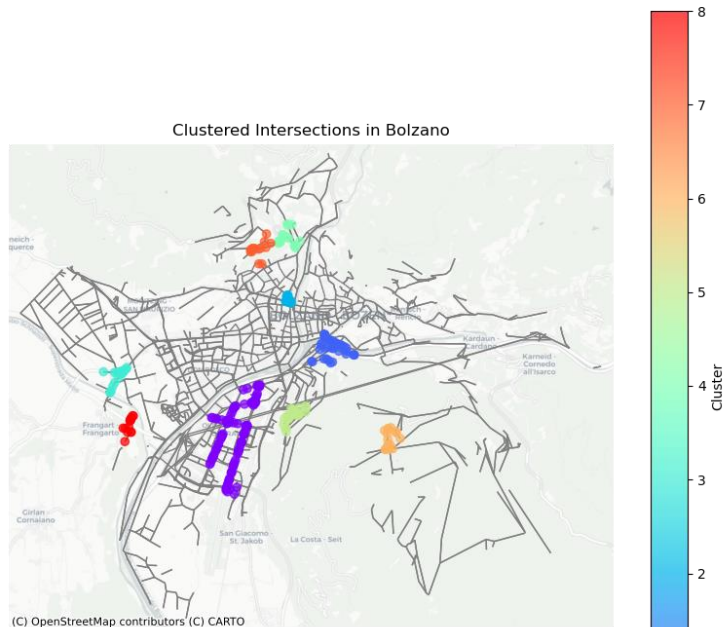
Silhouette Score: 0.54514
Davies-Bouldin Score: 0.54626
Calinski-Harabasz Score: 5227.40433



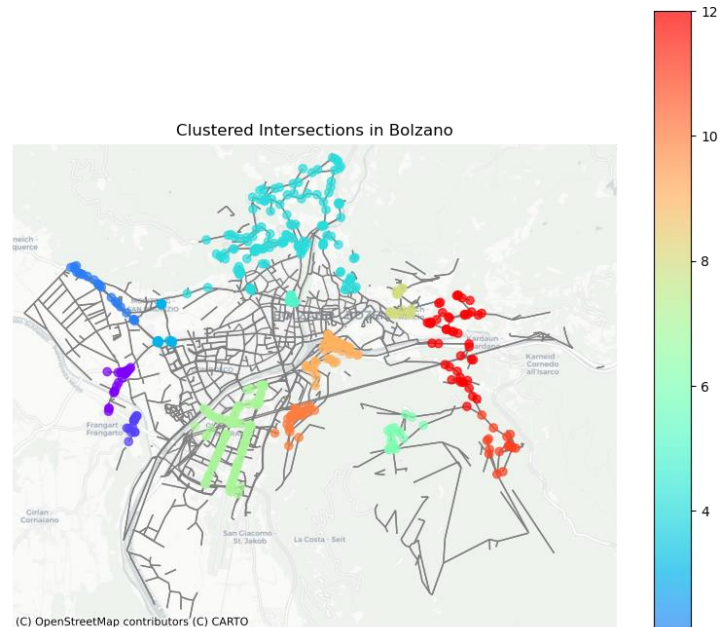
Silhouette Score: 0.68213
Davies-Bouldin Score: 0.48318
Calinski-Harabasz Score: 5451.08294

Results

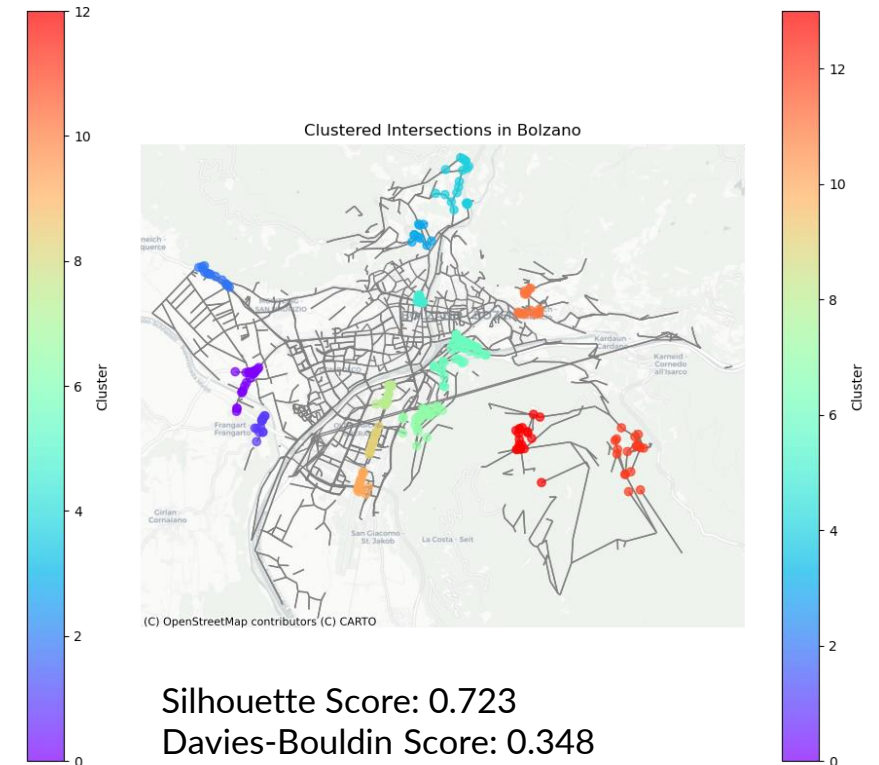
Infrastructure and Network Analysis (Bolzano) - Clustering High Stress (DBSCAN, HDBSCAN, OPTICS)



Silhouette Score: 0.652
Davies-Bouldin Score: 0.390
Calinski-Harabasz Score: 1690.83



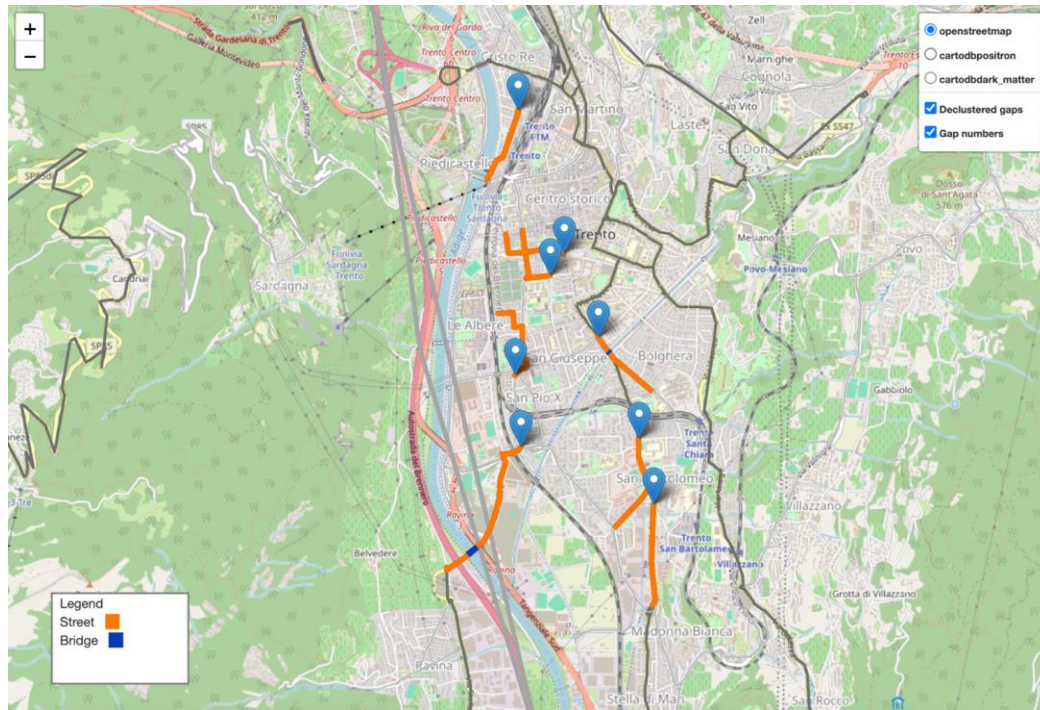
Silhouette Score: 0.442
Davies-Bouldin Score: 0.580
Calinski-Harabasz Score: 638.45



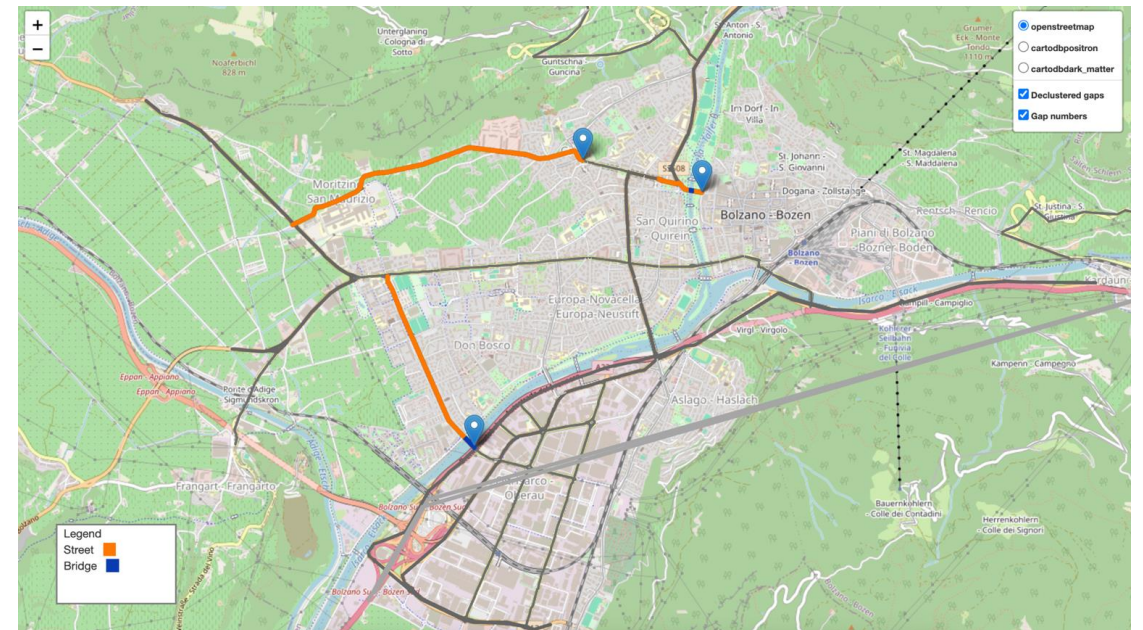
Silhouette Score: 0.723
Davies-Bouldin Score: 0.348
Calinski-Harabasz Score: 2371.376

Results

Infrastructure and Network Analysis - Gap Analysis (LTS IPDC procedure)



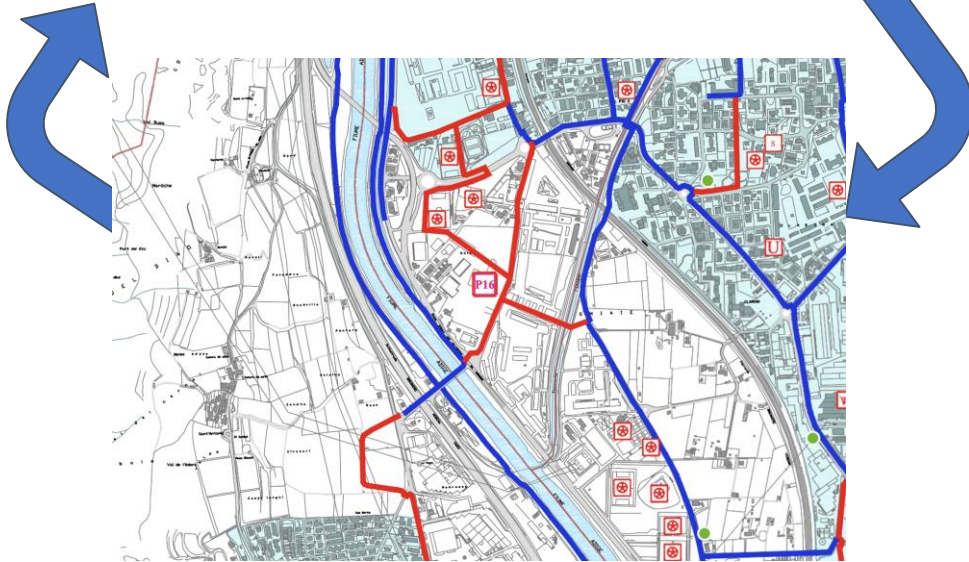
Trento



Bolzano

Results

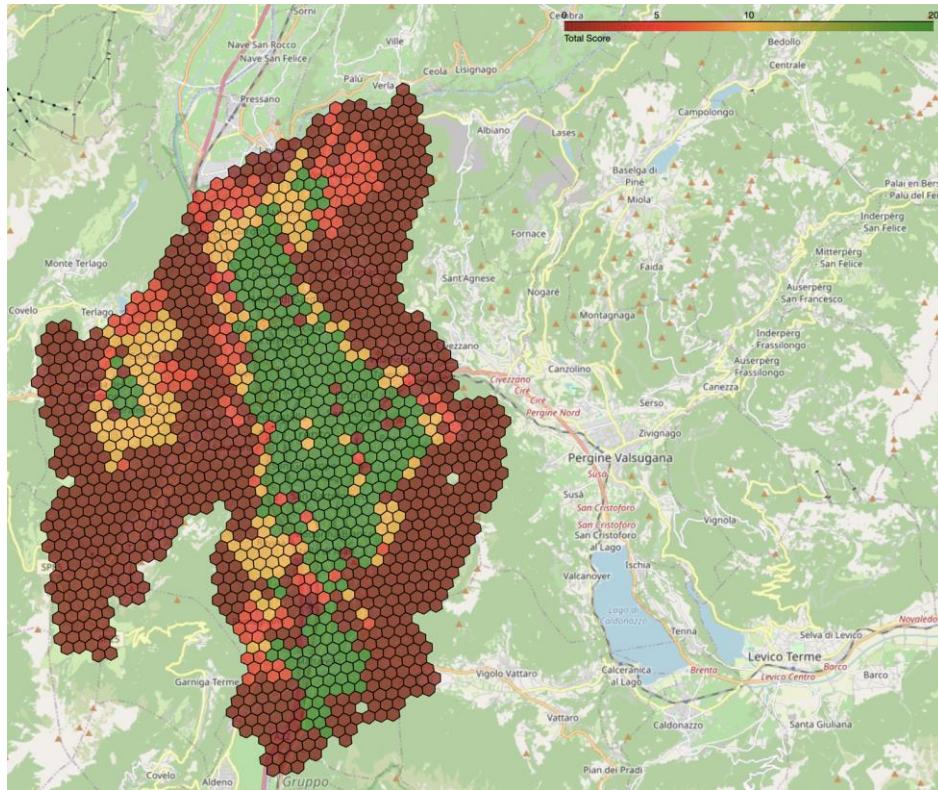
Focus on gap “Ravina Bridge” - Trento



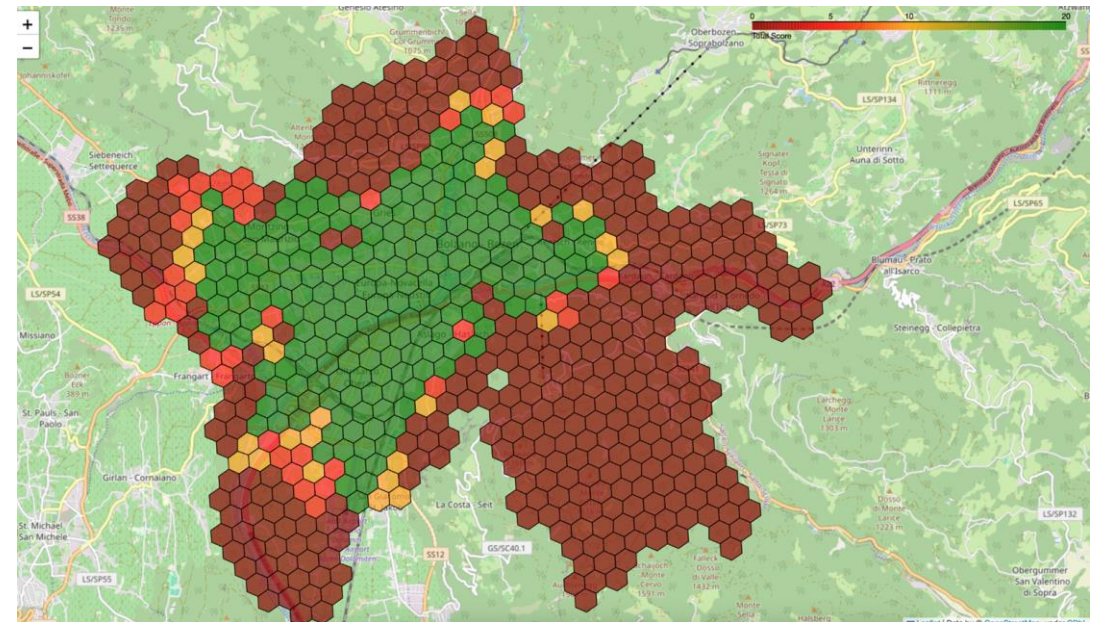
As can be seen from the image, the **Trento Biciplan** (photo below) **shows how the gap** highlighted by the LTS-IPDC analysis (top left photo) **is part of future plans to expand the bicycle network**, which is already in place near the bridge (as seen in the top right image where the red arrows indicate bicycle routes separated from the roadway).

Results

Infrastructure and Network Analysis - BNA Score (People for Bikes)



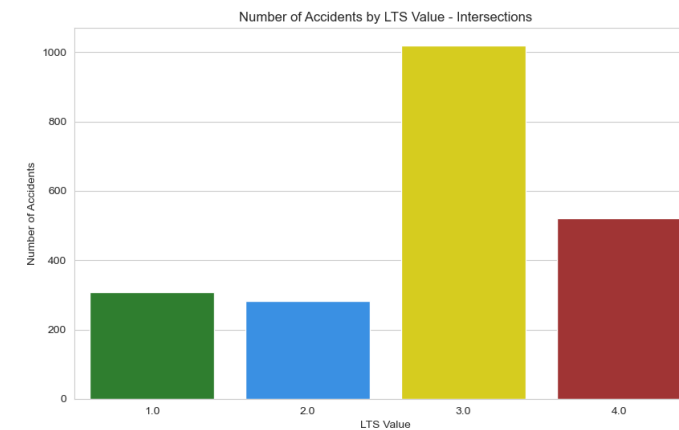
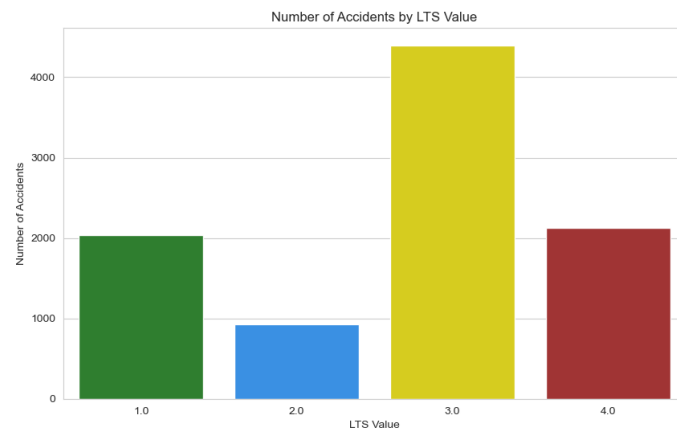
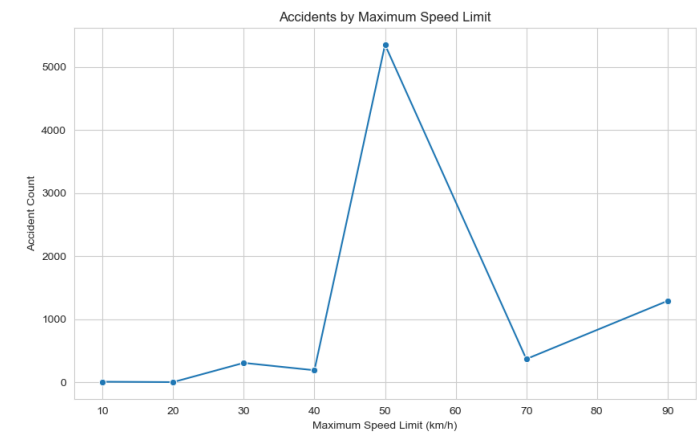
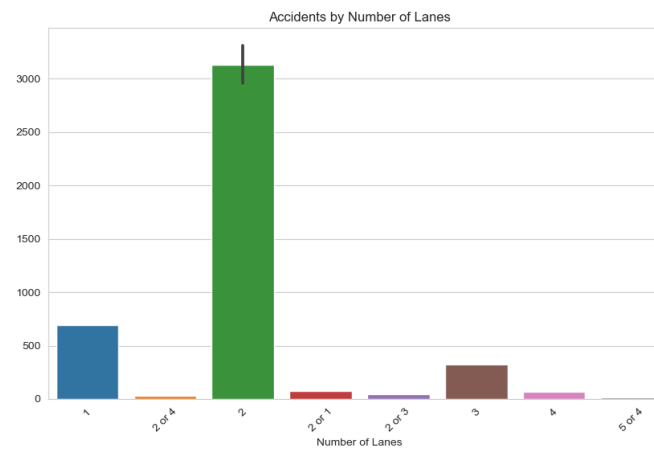
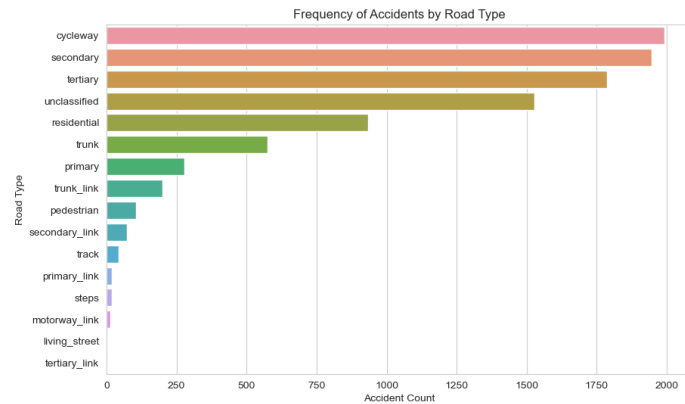
Trento



Bolzano

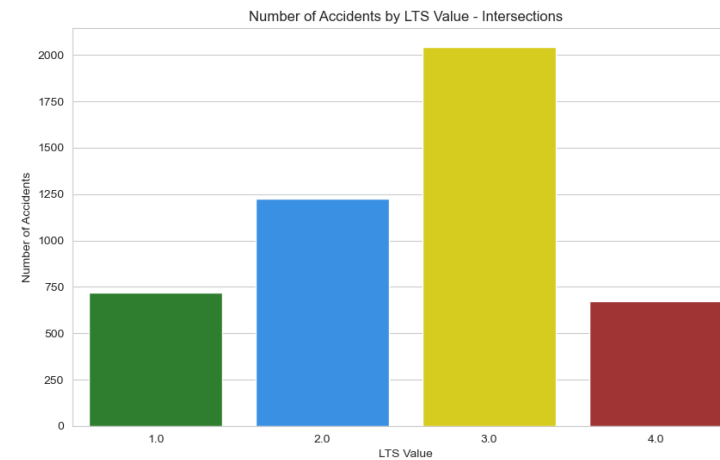
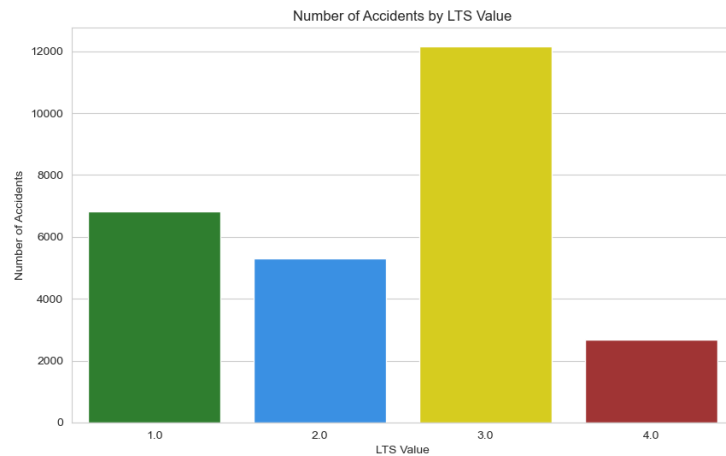
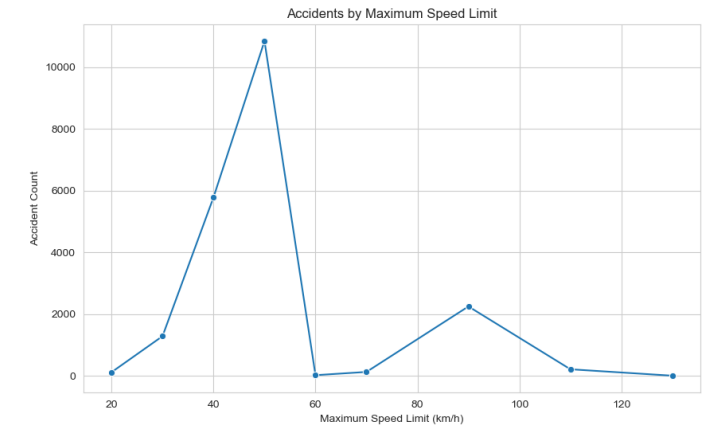
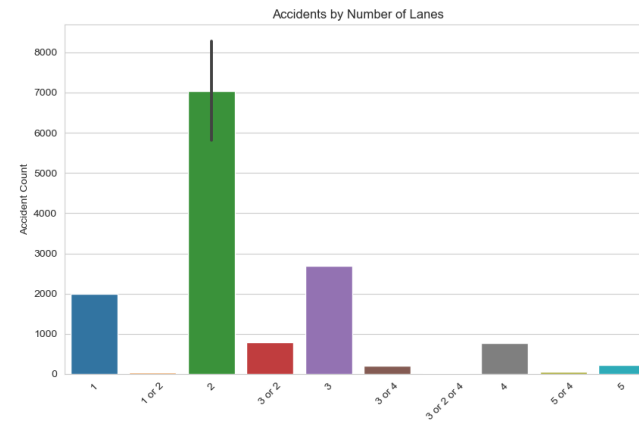
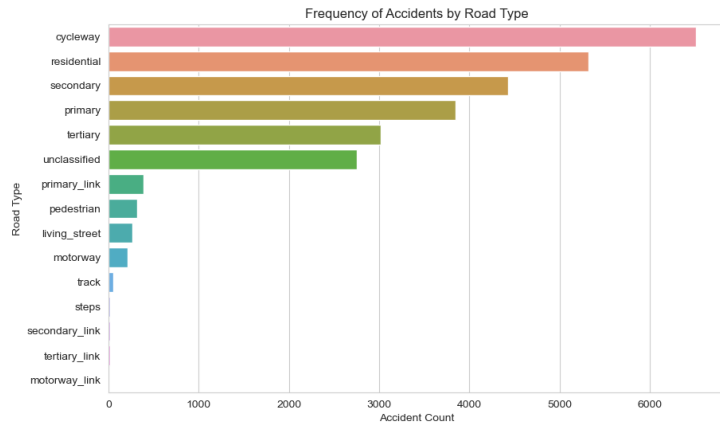
Results

Infrastructure and Network Analysis (Trento) - Accident Analysis



Results

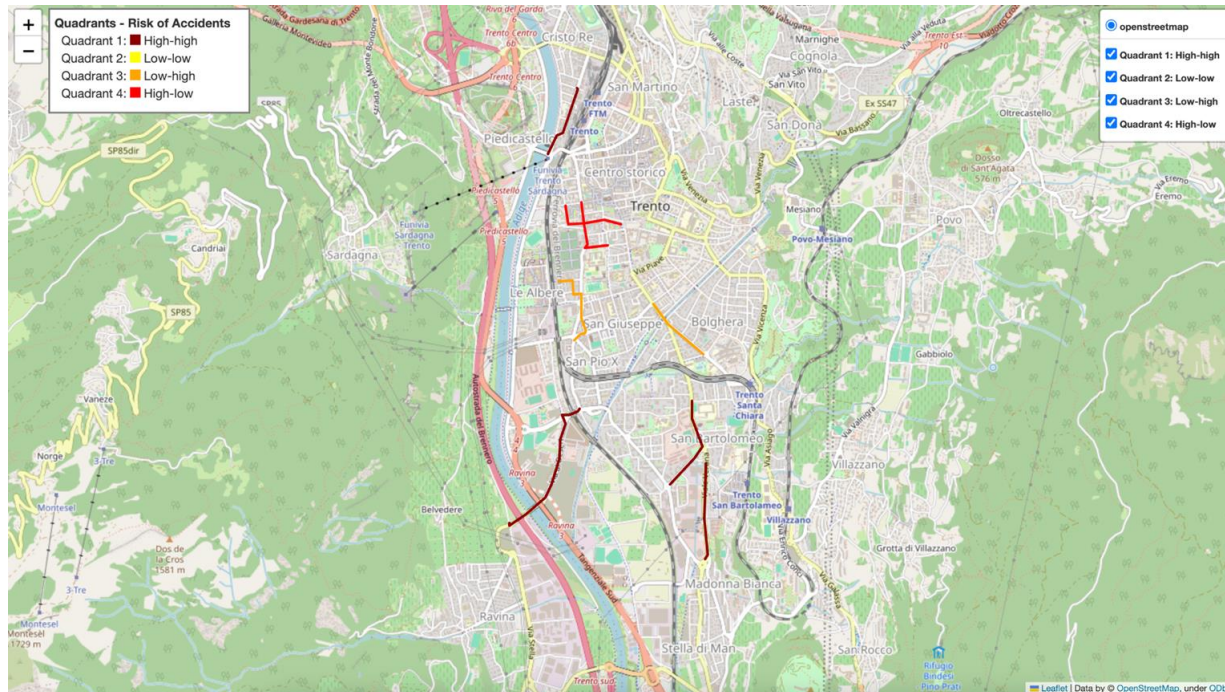
Infrastructure and Network Analysis (Bolzano) - Accident Analysis



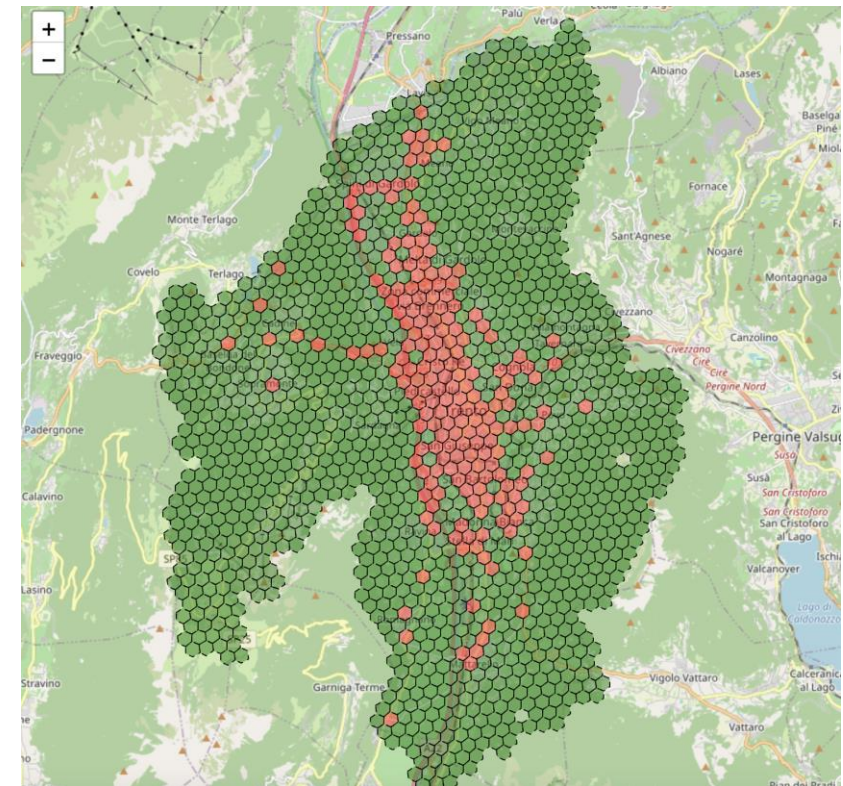
Results

Final Steps Analysis (Trento)

Spatial Relationship Analysis



High Risk Areas Prediction



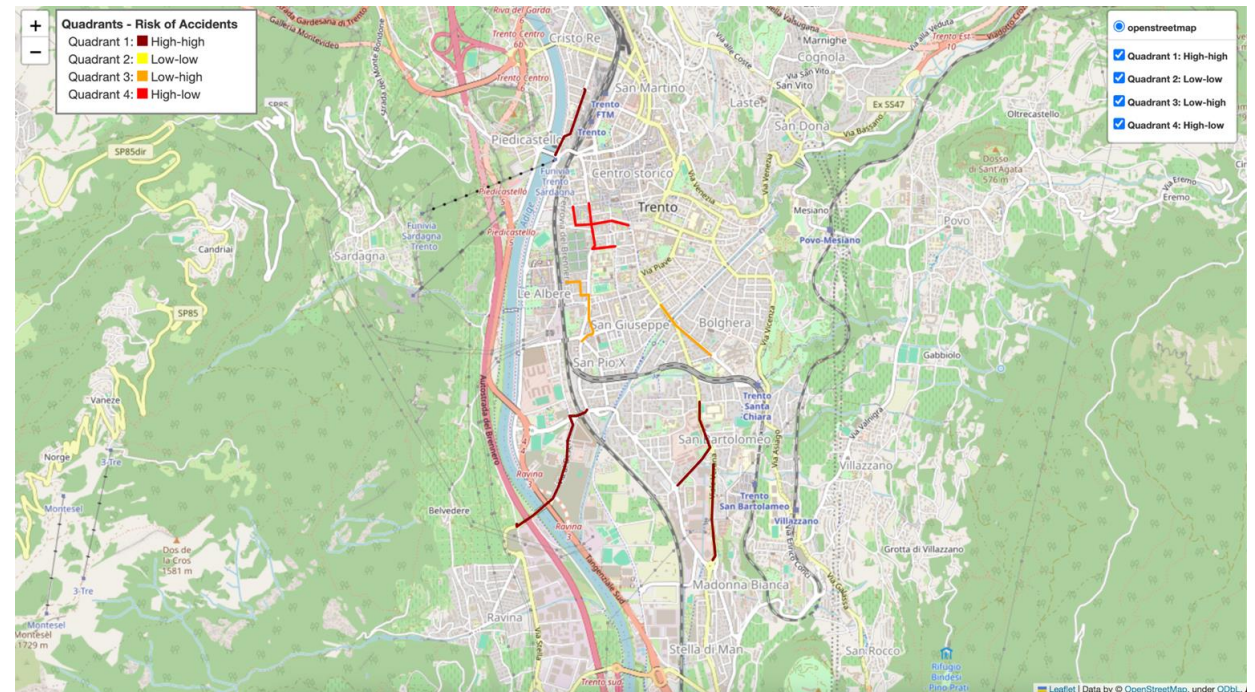
Results

Final Steps Analysis (Trento)

Risultati

Quadrant	Description	Number of Components
1	High-High (HH)	4
2	Low-Low (LL)	0
3	Low-High (LH)	2
4	High-Low (HL)	2

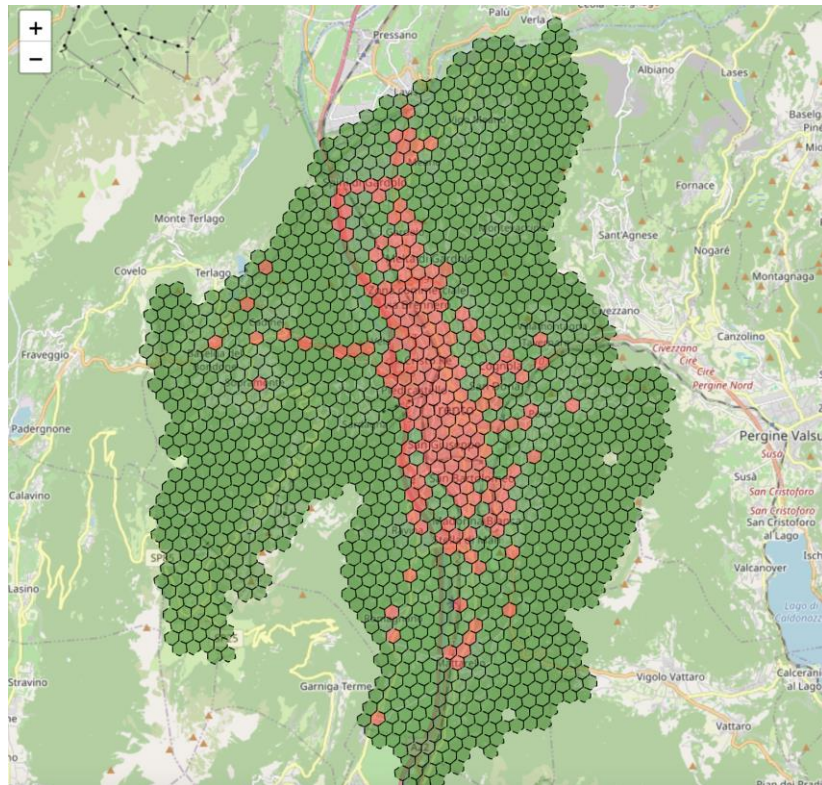
Spatial Relationship Analysis



Results

Final Steps Analysis (Trento)

High Risk Areas Prediction



Risultati

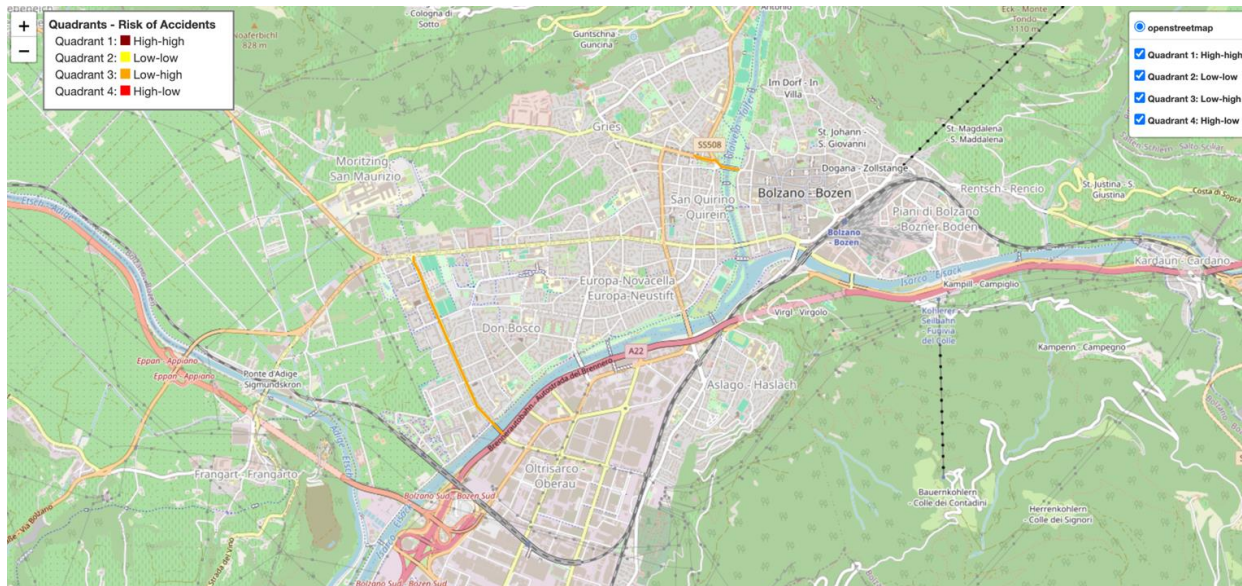
Metric	Logistic Regression	Random Forest Classifier	Support Vector Machine
Accuracy	0.66	0.89	0.71
Macro Avg F1-Score	0.59	0.79	0.63
Precision-Recall AUC	0.80	0.79	0.61
ROC-AUC Score	0.92	0.93	0.86

To have greater accuracy, keep balance between precision and recall I choose **RFC**.

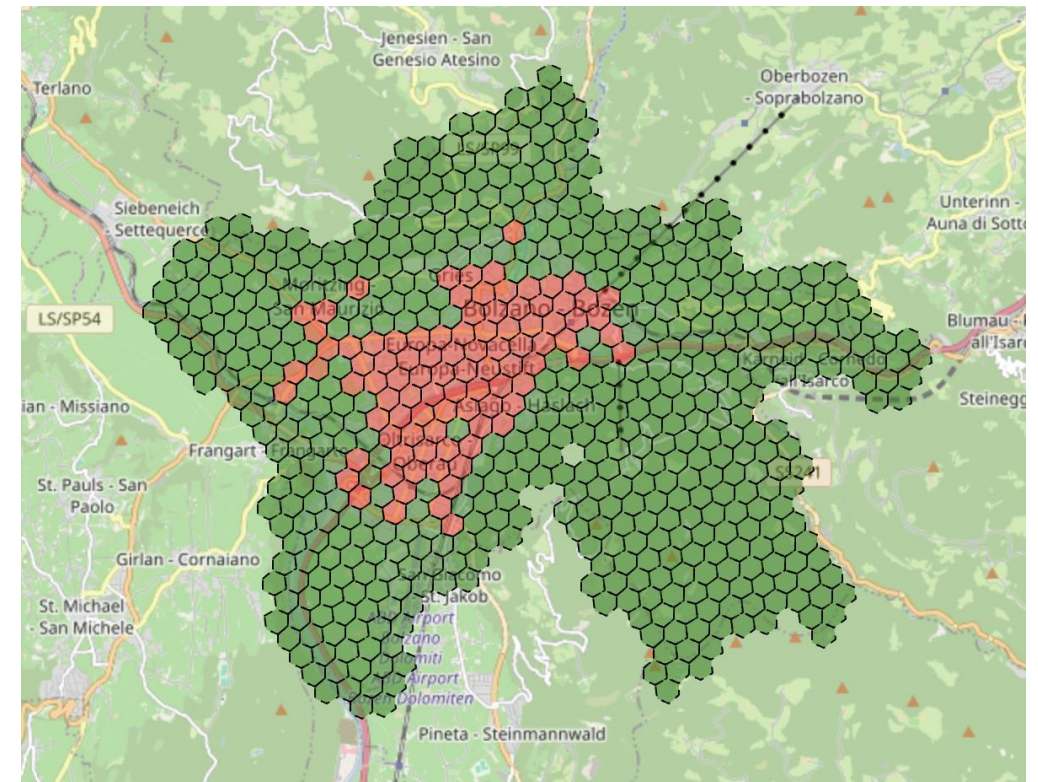
Results

Final Steps Analysis (Bolzano)

Spatial Relationship Analysis



High Risk Areas Prediction



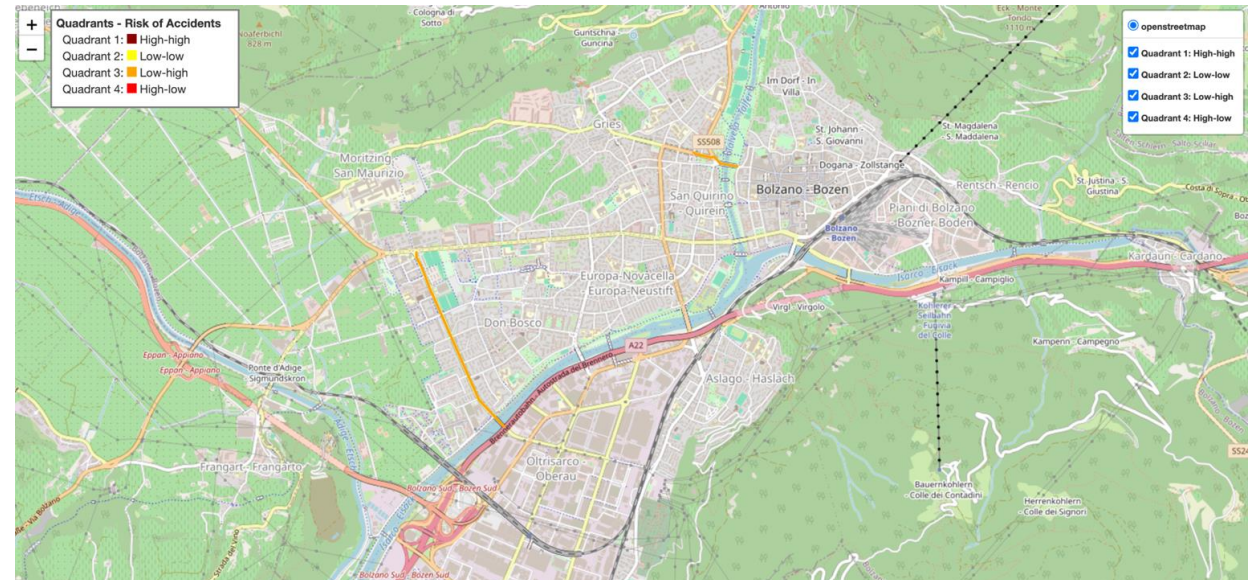
Results

Final Steps Analysis (Bolzano)

Risultati

Quadrant	Description	Number of Components
1	High-High (HH)	0
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4	High-Low (HL)	0

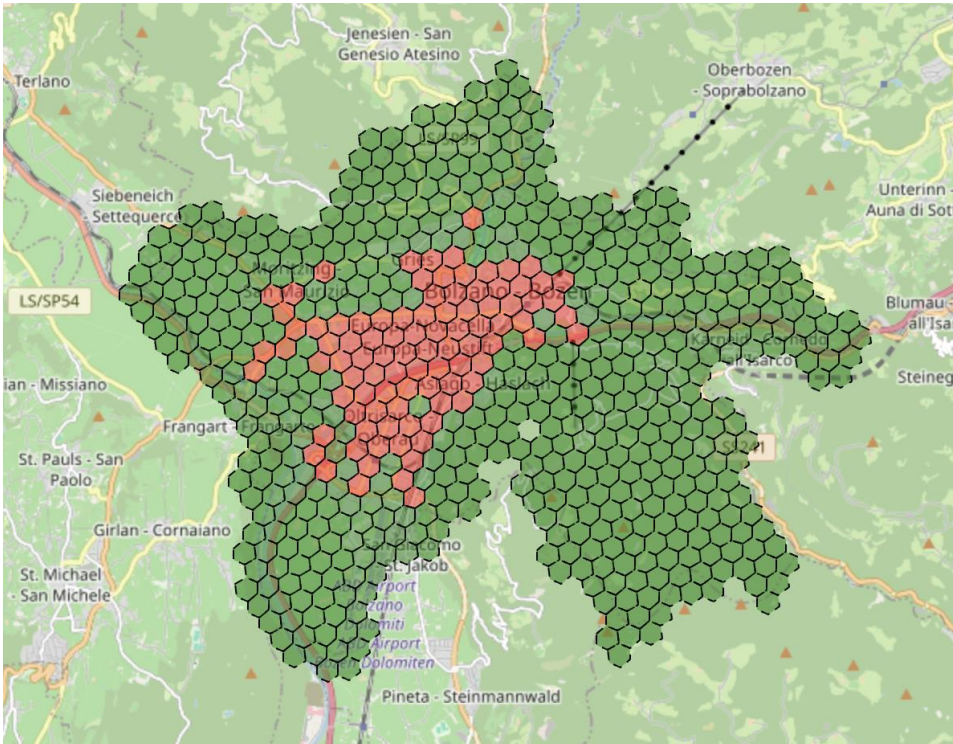
Spatial Relationship Analysis



Results

Final Steps Analysis (Bolzano)

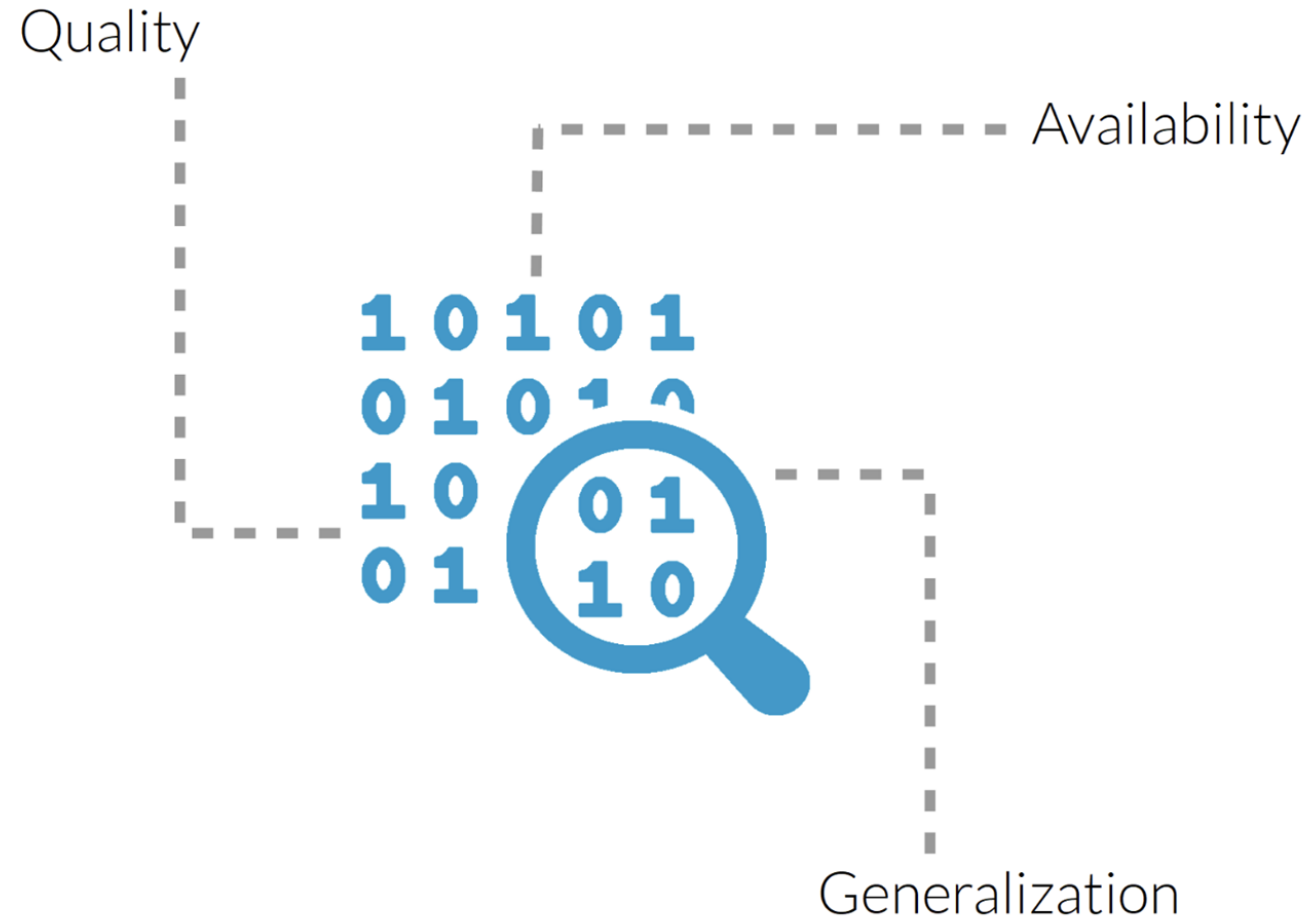
High Risk Areas Prediction



Risultati

Metric	Random Forest Classifier
Accuracy	0.92
Macro Avg F1-Score	0.91
Precision-Recall AUC	0.93
ROC-AUC Score	0.96

Limitations and Future Research



Conclusions

- 1) In urban landscapes with diverse geomorphological characteristics, **which streets or areas exhibit high LTS**, and which ones are characterized by low stress?
- 2) **Which streets and areas in the urban landscape exhibit higher BNA scores** and are in close proximity to common destinations, including services and amenities?
- 3) Are the networks composed of subgraphs distinguished by different levels of traffic stress? If this is the case, **how many high-stress gaps occur within different areas of low stress?** Moreover, which of these high-stress gaps are most **crucial**?
- 4) Is it possible to **predict areas at high risk for accidents** by combining quantitative metrics, such as centrality measures, with qualitative assessments like accessibility scores and levels of traffics stress?
- 5) Does a **spatial relationship** exist between the **locations of accidents and the high traffic stress gaps** within the urban network?

Thank you for your attention